

Sensorized Robotic Hugs with CASTOR Robot: Evaluation of Two Methods to Control Hug Physical-Interaction

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Abstract. Recently, the benefits and needs of human beings to maintain physical and social interactions have been reported. Thus, the consequences of the pandemic have been both physical and emotional. Social robots have been developed to maintain emotionally supportive physical contact such as hugging. The outcomes of the adaptation of the CASTOR robot to give and receive reciprocal sensor-controlled hugs in 12 participants (7 males, 5 females, 22.16 ± 2.08 years old, 1.67 ± 0.077 m of height) are presented. The results demonstrate the robot's ability to give comfortable reciprocal hugs to participants despite not being human-sized, and indicate that the CASTOR robot withstands physical interactions with adults.

Keywords: Hugs, Human-Robot Interaction, Social Robotics

1 Introduction

Physical contact is necessary for human beings to maintain psychological, emotional, and bodily well-being. It increases the hormones that produce the sensation of joy and happiness [10]. There is an evolution of physical contact needs, from childhood where complete physical contact is required to the development process to adulthood and elderly, where gestures of affection for emotional support are demanded [24], [17].

Hugging is the main sign of affection. It has different meanings, such as showing friendship, affection, gratitude, and emotional support [8]. Among the associated benefits are reducing stress and tension levels by releasing oxytocin, reducing the risk of dementia by giving tranquillity and confidence, reduction of blood pressure by activating Pacini corpuscle receptors in the skin [15]. Additionally, it increases self-esteem and improves mood by generating serotonin and dopamine release. Finally, stimulating oxygenation and strengthening the immune system [1].

However, due to the COVID-19 pandemic, this type of psycho-affective and physical support has been limited considerably in society [18], based on physical distancing and especially in population groups that significantly require these interactions, as in the older population [13]. Such is the need for physical contact in the pandemic that social interactions have changed and adapted to the current context [16]. This social limitation has led to increased levels of depression and anxiety [14]. Otherwise, stress and difficulty sleeping cases have increased too [6].

Recently, there has been an increase in research and projects aimed at developing social robots. One application for these social robots has been hugging, where some studies describe the design methods of hug robots [11], the evaluation of the emotional perception after hugging a robot [3], and the relationship of human-robot interaction with a stress measure [19], the relation of the perception with the release method implemented in the hug [2], and variables measured with sensors in the interaction [12].

Casas et al. [7] developed the CASTOR (CompliAnt Soft Robotics) robot (Center of Biomechatronics, Colombia) using compliant mechanisms. CASTOR is a low-cost open-source platform that integrates soft actuation to provide the necessary robustness and security for physical interaction. CASTOR was initially created to support therapies for children with Autism Spectrum Disorder (CwASD) and has been implemented in physical interaction studies with CwASD [20]. Recently, the robot was modified to assess its feasibility in hugging interactions by analyzing the perception in adults after interacting with the robot in four methods of hug release [9].

This study evaluates the interaction time during a hugging interaction through two sensors hugs (a touch sensor and a pressure sensor) and assess the adaptability of the CASTOR robot to be implemented in emotional support by physical human-robot interaction (pHRI). This paper is divided as follows: Section II presents related works in terms of robotic hugs research. Section III shows the methodology applied in this work with the CASTOR robot. Section IV and V show the results and the discussion of the variables evaluated. Finally, conclusions and future works are drawn in Section VI.

2 Related work

Block and Kuchenbecker [13] have studied the hug characteristics that affect the perception of a robotic hug [3]. They used the PR2 robot (Willow Garage, USA)

to evaluate physical variations of warmth and softness in the perception of hugs, where they also analyzed the pressure exerted by the participants. In addition, *Block et al.* [4] evaluate time variations, where 3 time variations were applied in the 3 physical variations, i.e. short time (1 second), average time (2.5 seconds) and long time (5 seconds) with the variations hard and cold, soft and cold and soft and warm. They found that other essential characteristics should be applied to a robot to obtain a quality hug [3] [4]. Recently, *Block et al.* [2] developed the Huggiebot 2.0 robot (MPI-IS and ETH, Germany and Switzerland). They evaluated other metrics to implement the robot in the hugging activity. Among them were human size, autonomous release methods, and visual perception of the participant. They found significant results in the perception of 5 metrics evaluated [2].

There are potential health benefits to interacting with a social robot. *Murphy et al.* [19] found that being hugged is associated with decreased negative effects of interpersonal conflict in adults [19]. Similarly, *Shiomi et al.* [21] found that hugging a robot increases people’s prosocial behaviours with a voluntary behaviour to help others [21].

Kim et al. [12] developed a huggable robotic plush toy for physical interaction with children. They quantified the range of pressure exerted by children when performing soft and hard hugs, involving 28 children aged 4 to 10 years who hugged the pressure-sensing tool. They found the maximum hug pressure of 2,623 psi. In this regard, they conclude that further analysis of the hugging data collected is needed [12]. Moreover, *Hedayati et al.* [11] developed a hugging study with children by modifying a 1-meter tall panda stuffed animal, adding 3D printed arms with actuators to perform the hugging action. Additionally, added an inner shirt composed of 8 pressure sensors.

Considering the above, this work implemented the CASTOR robot adapted to the social interaction of hugging to evaluate the time of two sensors-controlled hugs.

3 Methods

As previously stated, this work seeks to evaluate the participant’s time during a hugging interaction with the social assistance robot CASTOR through two sensors. The duration of the touch-controlled and the pressure-controlled hugs was measured through the graph obtained from the interaction. The touch and pressure sensor graphs show two events in the time, one at the beginning of the hug (start time) and one at the end of the hug (end time), from which the duration was obtained by subtracting the end time from the start time. In addition, the pressure sensor outcomes were analyzed for determine the maximum pressure supported by the CASTOR robot.

The sensor data were stored from the beginning to the end of all tests. These values were processed through MATLAB (MathWorks, 2018b, USA) with the lowpass (X , F_{pass} , F_s) filter function. Where F_s is the data’s sample rate in Hertz (Hz). F_{pass} is the passband frequency of the filter in Hz. Thus, the values

used for the filter are $(data, 1, 10)$. For the touch sensor, the hug time was measured from the valleys corresponding to the moment the user made contact with the sensor. For the pressure sensor, the time of the hug was measured from the peaks corresponding to the moment the user exerts a force on the balloon.

Using the *find peaks* function of MATLAB, the values of the peaks of the pressure sensor data were obtained. This information provided the maximum pressure peaks and the duration of the hug for each method (touch-sensor-controlled hug and pressure-sensor-controlled hug).

3.1 Robotic device

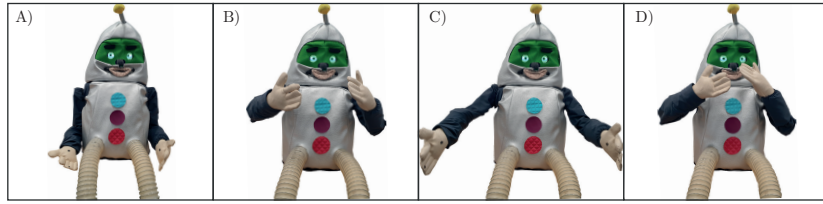


Fig. 1. CASTOR’s arms movements to perform a hug. A) Neutral position. B) Preparation position. C) Opening arms. D) Closing arms.

As reported previously, the CASTOR robot was modified for the study [9]. In the mechanical part, a change was made on the shoulder motors to lift the arms for a more extended period. Also, a pneumatic system composed of a balloon, an air pump, and a pressure sensor was added. Additionally, the touch sensors located at the back of the CASTOR’s head, made of Velostat (Adafruit, New York, USA) were used.

Consequently, four movements were established to perform a hug: a neutral position where the robot’s arms are close to the thorax, a preparation position to warn the participant to start the hug, and the opening of the arms to start the hug, finally, the closing of the arms, as shown in Figure 1.

In addition, two different methods to initiate and release the hug were performed. The first method controls the opening and closing of the arms is controlled using the touch sensor. In the second method, the opening and closing of the arms are controlled by exerting force on the balloon to increase the air pressure. Both methods can be seen in Figure 2.

3.2 Set-up

Considering that the height of CASTOR is 54 cm, a suitable height is required according to the participant’s height. A setup with three different heights was established, as shown in Figure 3. The height used for each participant corresponds to the next low closest to the participant’s height.

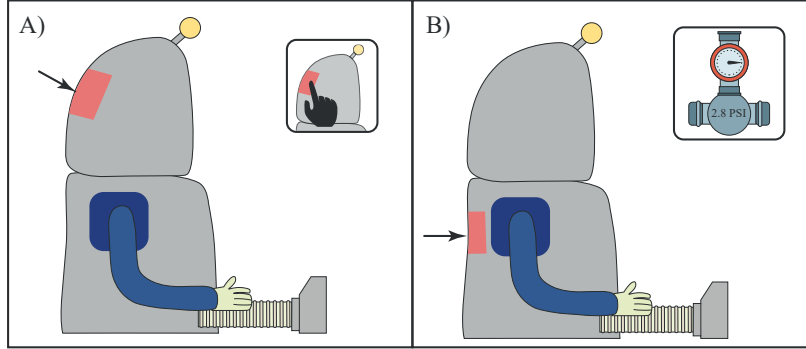


Fig. 2. Hugs implementation description. The arrow indicates the location of the sensor in each case. A) Touch sensor-controlled hug. B) Pressure controlled hug.

For the analysis of the participant's and the robot behaviour, the interventions of the 12 subjects in their two interactions with the robot were recorded using two cameras, one located behind the robot and the second located on the interaction side.

3.3 Participants

A total of 12 healthy subjects (7 males, 5 females, 22.16 ± 2.08 years old, 1.67 ± 0.077 m of height) performed the study. The study was approved by the Ethical Committee in the Colombian School of Engineering Julio Garavito. All subjects were informed about the scope and purpose of the experiment, and all participating individuals signed an informed consent form.

By easily adapting to participants' physical characteristics the setup was implemented depending on the participant's height. Table 1 shows the setup used for each participant according to the setups shown in Figure 3. As seen, no participant used the C set-up shown in Figure 3. However, it was available for the intervention.

Table 1. Corresponding setups for the 12 participants.

Participant	1	2	3	4	5	6	7	8	9	10	11	12
Setup	B	A	B	B	B	B	A	A	B	B	A	A

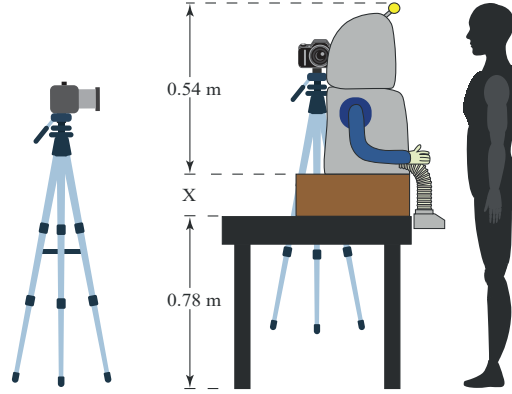


Fig. 3. X corresponds to different box sizes. Thus, the heights available for the participants are: A) The 1.49m height: With a 17cm box. B) The 1.63m height: With a 31cm box. C) The 1.80m height: With a 48cm box.

3.4 Procedure

The study is performed only once per participant and takes an estimated time between 15 to 25 minutes. In the beginning, the activity to be performed was explained, and the participants read and signed the informed consent form.

Then the participant is conducted inside the room, introduced to the robot, and allowed to interact with it. Next, both hugs were performed. First, the touch sensor-controlled hug, where the participant controls the opening and closing of the robot's arms using the touch sensor. And second, the air pressure controlled hug, where the participant must exert force on the balloon located on the robot's back to increase the air pressure (above 2.8 psi), and thus control the opening and closing of the arms. Both hugs are described in Figure 2.

After the hugs, a space for comments is given to the participants to talk about the experience with the robot and the specific opinions of the interactions.

3.5 Variables

The main variable is the interaction time in each of the two scenarios. This variable is taken from the combination of the information obtained from the sensors and the rectification in the video analysis. Finally, the perception of the social robotics and each of the hugs (including two time-controlled hugs) was previously acquired and reported [9].

Statistical analysis is performed with the obtained results. For this purpose, the mean and standard deviation of the time of the touch-sensor controlled hug and the time of the pressure sensor-controlled hug are calculated with the values obtained by all the participants. In addition, the time and pressure results of

the subjects are qualitatively compared with each of the heights available in the set-up, to analyze the adaptability of the CASTOR robot to the varying height characteristics of the subjects.

4 Results

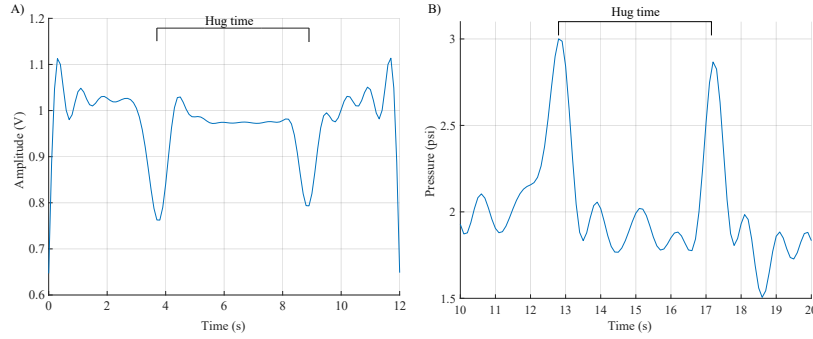


Fig. 4. Filtered examples obtained from sensors. A) Touch sensor in hugging. B) Pressure sensor in hugging.

All 12 participants completed the study successfully. A graph was obtained for each participant for the touch sensor (Figure 4A) and the pressure sensor (Figure 4B). Hence, as mentioned in Section III and considering the peaks and valleys showed in the Figure 4, the values of interaction time were determined. In addition, for the pressure sensor, the pressure peaks and the peak-to-peak mean were obtained as shown in the Figure 4B. Otherwise, due to the different local peaks near to the amplitude of the start-end-peaks, in some participants the use of the videos was necessary for determine which peak corresponds to the beginning and which to the end of the hug. Table 2 shows the study results for both types of methods, touch-sensor controlled hug and pressure-sensor controlled hug.

Finally, the mean and standard deviation for the times of the touch sensor-controlled hug and the pressure sensor-controlled hug of all participants were obtained with a value of 7.83 ± 4.83 s and 7.70 ± 4.43 s, respectively. The values indicate similar time values in both cases, however, a high standard deviation is denoted and will be discussed as follows.

5 Discussion

The CASTOR robot has had physical human-robot interaction (pHRI) evaluations previously analyzed in treatment scenarios in CwASD [20], where the

ID	Touch sensor- controlled hug	Pressure sensor- controlled hug			
	Time (s)	Initial pressure peak (psi)	Final pressure peak (psi)	Mean pressure (psi)	Time (s)
1	1.8	2.98	3.15	2.04	4.5
2	5.1	2.78	3.42	2.39	1.6
3	9.5	3.12	3.17	2.68	2.1
4	4.4	3.52	4.27	1.85	14.0
5	2.9	4.60	5.48	2.65	5.6
6	4.7	3.62	3.58	2.15	5.2
7	5.1	3.73	3.88	3.16	4.6
8	13.1	4.45	4.06	2.37	11.8
9	14.9	3.61	3.19	2.13	13.1
10	5.6	4.70	4.05	2.26	6.8
11	15.8	4.17	4.25	1.79	11.2
12	11.0	3.19	3.14	1.82	11.9

Table 2. Time and pressure outcomes for the 12 participants.

importance of maintaining these interactions in therapy environments has been tested and evaluated. Moreover, its implementation in adult population was evaluated through an evaluation study of 4 different hugs, two time-controlled and two sensor-controlled, with perception questionnaires [9]. The interpretation provided by the sensors in the control of the action is essential to analyze different qualities of the hug, which can help to characterize and improve the hug interaction with the CASTOR robot. In this case, as the hugs are sensor-controlled, the interactions become more natural, which is why they are rated as more pleasant for the participants. However, considering several factors reported by *Block et al.* [2] for control the huggable robot, CASTOR still lacks. Therefore, it should be considered to implement more accurate and sensitive sensors to generate a behavioral control of the robot against the variables obtained from the environment and users, likewise, generate a more pleasant interaction for users.

Ergonomic robots designed for the hugging action have been developed, such as *Block et al.* [2], [5] who designed the Huggiebot 2.0 and huggiebot 3.0 social robots. They found and evaluated metrics for hugging, including size and human-like physical appearance. Due to the fact that hugging is an action anatomically adapted by humans to generate mutual physical contact and generate a sense of well-being [10]. However, the results of the present study demonstrate that the CASTOR robot, which is not human-sized, was able to give comfortable reciprocal hugs for all the volunteers. *Block et al.* [3] performed an evaluation of three different hugs evaluating characteristics such as softness and warmth in the hugging interaction, where they found that the soft-warm hugs were more comfortable and safety than the hard-cold hugs, through perception evaluations, being essential variables to consider when choosing or designing a robot to be huggable. Likewise, the CASTOR robot is soft and spongy to the touch, has a huggable system based on compliant soft Robotics with internal pistons that counts as a passive degree of freedom to avoid structural and functional damages of the robot [7], being a robot made from participatory design with robot

appearance but bio-inspired in the human being (anatomical distribution), and being an adaptable and low-cost robot.

In the present study the adaptability of the CASTOR robot to provide comfortable reciprocal hugs for participants of different heights wanted to be evaluated. However, due to the lack of volunteers with different anthropometric characteristics within the three set-up options, this adaptability could not be evaluated, which should be evaluated in a future study with a varied and controlled group of subjects. Furthermore, *Block et al.* [2] conducted an investigation of hug interaction reporting that immediate-sensor-controlled hugs received better perception scores than time-controlled hugs, in the same manner as the perception assessment in the CASTOR robot previously analyzed [9]. However, the mean time obtained here in the two sensor-controlled hugs analyzed, are higher than the long-time hug reported by them, with means of 7.83 ± 4.83 s and 7.70 ± 4.43 s for the touch and pressure sensors, respectively.

Additionally, considering the data obtained, it is observed that the hugging time varies with each participant. Times of 15.8 s and 14.0 s were reached for the touch sensor-controlled hug and the pressure sensor-controlled hug, respectively. This is confirmed by *Shiomi et al.* [22] who found that people interacted with a robot for a longer time when the robot gives reciprocal hugs. Moreover, they found a greater robotic perception in robot hugs when the participant can control the interaction time compared to the time-controlled hugs. Also, the difference identified in the interaction times (high standard deviation) could be due to the appearance of the robot itself. *Trovato et al.* [23] concluded that appearance and factors such as noise and vibration, influence the participants' behaviour when interacting with a robot.

Considering the results of pressures obtained with a maximum pressure of 5.48 psi exerted on the CASTOR robot was analyzed, and the mean pressure was 2.27 ± 0.41 psi, higher than that reported by *Kim et al.* [12], who performed a study in children aged between 4 and 10 years, where the maximum pressure exerted was of 2.62 psi; however, the pressure exerted by a child can not be compared with the force of an adult. In addition, *Block et al.* [5], reported a study in adults based on a pressure sensor and a microphone, by interacting with hugs with the huggiebot 3.0; they took the sensors data for control the robot behaviour, instead of used it to analyze the interaction.

Despite the inclusion of two sensors to improve the physical interaction with the CASTOR robot, some limitations have been recognized. One limitation is the movement performed by the robot, that requires improvement to be more natural. This is possible through control and machine learning algorithms based on human motion, improving the interaction as *Block et al.* [5], who evaluated the perception of the subjects with a scale from 0 to 10 of 16 hugs, evaluating 4 different types of hugs and the combination between them considering the response of the Huggiebot 3.0 robot controlled by a pressure sensor, and the detection of the user (staying still, moving the hand vertically on the user's back, tapping on the user's back, or tightening hold the user), depending on the type of hug given by the participant (hold still, rub the robot's back, pat the

robot’s back, or squeeze the robot). In addition, they included a microphone and subject data to develop a perceptual algorithm that classifies the user’s actions. Another limitation is the lack of voice feedback essential for improve the interaction, and due to some users claimed to include a voice line to indicate the moment when the robot is ready to interact with it.

6 Conclusions and Future Works

Although the CASTOR robot was designed for a therapeutic purpose in CwASD. On the one hand, the study demonstrated how the robot withstands physical interactions with adults with pressures around 2.27 ± 0.41 psi and a maximum peak of 5.48 psi. On the other hand, although the interaction time of each participant was different, the inclusion of the 2 sensors evidenced that the CASTOR robot was able to give comfortable reciprocal hugs to the participants despite not being human-sized. However, the adaptability of the CASTOR robot at different heights still needs to be evaluated due to the lack of participants with different anthropometric characteristics within the three configuration options.

Future work will continue with improving the physical interaction of the CASTOR robot through the inclusion of voice lines and sensors to increase the variables obtained during the interaction. In addition, to include control and machine learning algorithms to improve the robot’s ability to achieve a more natural hug. Finally, to carry out the study again including a variety of participants in order to evaluate the three configuration options.

Conflict of Interest

The authors declare that they have no conflict of interest.

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